

# Commutative Algebra 1

## 1 Tensoring with $A/I$ : detailed explanation of the four examples

The common principle behind all four examples is the canonical isomorphism

$$(A/I) \otimes_A M \cong M/IM,$$

where:

- $A$  is a ring,
- $I \subseteq A$  is an ideal,
- $M$  is an  $A$ -module,
- $IM$  is the submodule generated by all products  $im$  with  $i \in I$  and  $m \in M$ .

Intuitively, tensoring with  $A/I$  means that every element of the ideal  $I$  is forced to act as 0. In other words, we are “reducing the module  $M$  modulo  $I$ .”

### 1.1 Element-level meaning of $(A/I) \otimes_A M$

An element of  $(A/I) \otimes_A M$  is a finite sum of pure tensors

$$(\bar{a}) \otimes m,$$

where  $\bar{a} = a + I \in A/I$  and  $m \in M$ .

Because  $\bar{a} = a \cdot \bar{1}$  in  $A/I$ , we have

$$(\bar{a}) \otimes m = (\bar{1}) \otimes am.$$

So every pure tensor can be rewritten in the form

$$\bar{1} \otimes m.$$

Now if  $i \in I$ , then  $\bar{i} = 0$  in  $A/I$ , hence

$$\bar{1} \otimes (im) = \bar{i} \otimes m = 0.$$

Therefore every element of  $IM$  becomes zero in the tensor product.

This is exactly why the tensor product depends only on the class of  $m$  modulo  $IM$ . The isomorphism is:

$$\Phi : (A/I) \otimes_A M \longrightarrow M/IM, \quad (\bar{a}) \otimes m \longmapsto am + IM.$$

Its inverse is given by

$$\Psi : M/IM \longrightarrow (A/I) \otimes_A M, \quad m + IM \longmapsto \bar{1} \otimes m.$$

So the tensor product remembers precisely the part of  $M$  that survives after we impose the rule  $I = 0$ .

## 1.2 What survives and what dies

Under the passage from  $M$  to  $(A/I) \otimes_A M$ :

- terms not involving the ideal  $I$  survive,
- all terms in the submodule  $IM$  vanish.

Thus:

- over  $\mathbb{Z}$ , tensoring with  $\mathbb{Z}/m$  means reducing modulo  $m$ ,
- over  $k[x]$ , tensoring with  $k[x]/(x)$  means setting  $x = 0$ .

## 2 Example 1: $A = \mathbb{Z}$ , $I = (m)$ , $M = \mathbb{Z}^r$

We compute

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r.$$

By the general formula,

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \cong \mathbb{Z}^r / m\mathbb{Z}^r.$$

Now

$$m\mathbb{Z}^r = \{(ma_1, \dots, ma_r) : a_1, \dots, a_r \in \mathbb{Z}\}.$$

So  $\mathbb{Z}^r / m\mathbb{Z}^r$  is obtained by reducing each coordinate modulo  $m$ . Therefore

$$\mathbb{Z}^r / m\mathbb{Z}^r \cong (\mathbb{Z}/m)^r.$$

Hence

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \cong (\mathbb{Z}/m)^r.$$

### 2.1 Element-wise description

Let  $e_1, \dots, e_r$  be the standard basis of  $\mathbb{Z}^r$ . Any vector  $v \in \mathbb{Z}^r$  can be written as

$$v = a_1 e_1 + \dots + a_r e_r.$$

Then

$$[1]_m \otimes v = [1]_m \otimes (a_1 e_1 + \dots + a_r e_r) = [a_1]_m \otimes e_1 + \dots + [a_r]_m \otimes e_r.$$

So the tensor product simply remembers the coefficients modulo  $m$ .

### 2.2 Concrete example

Take  $m = 5$ ,  $r = 2$ , and  $v = (7, 11)$ . Then

$$(7, 11) \equiv (2, 1) \pmod{5}.$$

Thus

$$[1]_5 \otimes (7, 11) \text{ corresponds to } ([2]_5, [1]_5) \in (\mathbb{Z}/5)^2.$$

## 2.3 Interpretation

The “vectors” here are genuine integer vectors in  $\mathbb{Z}^r$ , but after tensoring with  $\mathbb{Z}/m$ , only their coordinates modulo  $m$  remain visible.

## 3 Example 2: $A = \mathbb{Z}$ , $I = (m)$ , $M = \mathbb{Z}/n$

We compute

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n).$$

Using the same principle,

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \cong (\mathbb{Z}/n)/m(\mathbb{Z}/n).$$

So we must understand the subgroup  $m(\mathbb{Z}/n)$ .

### 3.1 Computing $m(\mathbb{Z}/n)$

Inside  $\mathbb{Z}/n$ , multiplication by  $m$  gives the subgroup

$$m(\mathbb{Z}/n) = \{[mk]_n : k \in \mathbb{Z}\}.$$

This is the cyclic subgroup generated by  $[m]_n$ .

The order of  $[m]_n$  in  $\mathbb{Z}/n$  is

$$\frac{n}{\gcd(m, n)},$$

so the subgroup  $m(\mathbb{Z}/n)$  has exactly that many elements.

Therefore the quotient  $(\mathbb{Z}/n)/m(\mathbb{Z}/n)$  has cardinality

$$\frac{n}{n/\gcd(m, n)} = \gcd(m, n).$$

Since the quotient of a cyclic group is cyclic, we get

$$(\mathbb{Z}/n)/m(\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

Hence

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

### 3.2 Element-wise description

The tensor product is generated by the single element

$$[1]_m \otimes [1]_n.$$

Indeed, for any  $a, b \in \mathbb{Z}$ ,

$$[a]_m \otimes [b]_n = ab([1]_m \otimes [1]_n).$$

So the whole tensor product is cyclic.

Moreover,

$$m([1]_m \otimes [1]_n) = 0 \quad \text{and} \quad n([1]_m \otimes [1]_n) = 0,$$

so the order of the generator divides both  $m$  and  $n$ , hence divides  $\gcd(m, n)$ . In fact, its exact order is  $\gcd(m, n)$ .

### 3.3 Concrete example

Let  $m = 12$  and  $n = 18$ . Then

$$\gcd(12, 18) = 6,$$

so

$$(\mathbb{Z}/12) \otimes_{\mathbb{Z}} (\mathbb{Z}/18) \cong \mathbb{Z}/6.$$

The element

$$[1]_{12} \otimes [1]_{18}$$

is a generator of this cyclic group.

### 3.4 Interpretation

This tensor product retains only the *common torsion* visible in both cyclic modules:

- $\mathbb{Z}/m$  has  $m$ -torsion,
- $\mathbb{Z}/n$  has  $n$ -torsion,
- their tensor product detects the part compatible with both, namely  $\gcd(m, n)$ -torsion.

## 4 Example 3: $A = k[x]$ , $I = (x)$ — evaluation at $x = 0$

Now let

$$A = k[x], \quad I = (x).$$

Then

$$A/I = k[x]/(x).$$

But quotienting by  $(x)$  kills all multiples of  $x$ , so every polynomial becomes equal to its constant term. Thus

$$k[x]/(x) \cong k,$$

via the map

$$f(x) \mapsto f(0).$$

So in this situation,

$$A/I \cong k.$$

Therefore for any  $A$ -module  $M$ ,

$$k \otimes_{k[x]} M \cong M/xM.$$

### 4.1 Why this means “set $x = 0$ ”

Since  $x = 0$  in the quotient ring  $k[x]/(x)$ , we have in the tensor product:

$$1 \otimes (xm) = x \otimes m = 0.$$

So every element of the form  $xm$  vanishes. That is why we divide by  $xM$ .

This is exactly the module-theoretic meaning of evaluating at  $x = 0$ : all  $x$ -terms disappear.

## 4.2 Particular case: $M = k[x]$

If  $M = A = k[x]$ , then

$$k \otimes_{k[x]} k[x] \cong k[x]/xk[x].$$

But  $xk[x]$  is the set of polynomials with zero constant term, so the quotient keeps only the constant term:

$$k[x]/xk[x] \cong k.$$

Thus

$$k \otimes_{k[x]} k[x] \cong k.$$

## 4.3 Element-wise example

Take the polynomial

$$f(x) = 3 + 2x + x^2.$$

Then in the tensor product,

$$1 \otimes f(x) = 1 \otimes (3 + 2x + x^2).$$

Since all terms involving  $x$  vanish,

$$1 \otimes (3 + 2x + x^2) = 1 \otimes 3.$$

So only the constant term remains.

## 4.4 Interpretation

Tensoring with  $k \cong k[x]/(x)$  means:

- forget all multiples of  $x$ ,
- keep only what survives at  $x = 0$ ,
- algebraically: pass from  $M$  to the quotient  $M/xM$ .

## 5 Example 4: $A = k[x]$ , $I = (x)$ , $M = A/(x^2)$

Now let

$$A = k[x], \quad I = (x), \quad M = A/(x^2) = k[x]/(x^2).$$

We want to compute

$$(A/I) \otimes_A M = k \otimes_{k[x]} A/(x^2).$$

By the general rule,

$$(A/I) \otimes_A M \cong M/xM.$$

So we must compute  $xM$ .

### 5.1 Structure of $M = A/(x^2)$

In the quotient  $k[x]/(x^2)$ , every polynomial is equivalent to one of the form

$$a + bx \pmod{x^2}.$$

So as a  $k$ -vector space,

$$M = k \cdot 1 \oplus k \cdot x.$$

A convenient basis is

$$\{1, x\}.$$

However, inside this module we have

$$x^2 = 0.$$

So  $x$  is not zero, but it is nilpotent.

### 5.2 Computing $xM$

Take an arbitrary element

$$a + bx \in M.$$

Multiplying by  $x$ , we get

$$x(a + bx) = ax + bx^2.$$

Since  $x^2 = 0$  in  $M$ , this becomes

$$x(a + bx) = ax.$$

Therefore

$$xM = kx.$$

So the quotient is

$$M/xM = (k \cdot 1 \oplus k \cdot x)/(k \cdot x) \cong k.$$

Hence

$$k \otimes_{k[x]} A/(x^2) \cong M/xM \cong k.$$

### 5.3 Element-wise tensor computation

Take the class of  $a + bx$  in  $M = A/(x^2)$ . Then in the tensor product,

$$1 \otimes (a + bx) = 1 \otimes a + 1 \otimes bx.$$

But

$$1 \otimes bx = b(1 \otimes x).$$

Now  $x = 0$  in  $k = A/(x)$ , so

$$1 \otimes x = x \otimes 1 = 0.$$

Therefore

$$1 \otimes (a + bx) = 1 \otimes a.$$

So again, only the constant part survives.

## 5.4 Interpretation

This example is important because  $M = A/(x^2)$  is larger than  $k$ : it has an extra “infinitesimal” direction  $kx$ . But after tensoring with  $k = A/(x)$ , the  $x$ -part disappears.

So even though  $x \neq 0$  in  $M$ , it becomes invisible after imposing  $x = 0$ .

## 6 What are the “vectors” or “elements” in these examples?

These are not always vectors in the geometric sense. They are simply *elements of modules*.

- In Example 1, the elements are tuples

$$(a_1, \dots, a_r) \in \mathbb{Z}^r.$$

These behave like integer vectors.

- In Example 2, the elements are residue classes

$$[a]_n \in \mathbb{Z}/n.$$

- In Example 3, the elements are arbitrary module elements  $m \in M$ , where  $M$  is any  $k[x]$ -module.
- In Example 4, the elements are classes

$$a + bx \pmod{x^2},$$

so each element has a constant part and an  $x$ -part.

In all cases, after tensoring with  $A/I$ , every element can be represented by something of the form

$$\bar{1} \otimes m,$$

and two elements become equal precisely when they differ by something from  $IM$ .

Thus the tensor product is measuring the module  $M$  after “forgetting” everything killed by the ideal  $I$ .

## 7 One-line intuition for each example

- **Example 1:** reduce each coordinate modulo  $m$ .
- **Example 2:** keep only the common torsion shared by  $\mathbb{Z}/m$  and  $\mathbb{Z}/n$ .
- **Example 3:** set  $x = 0$  in the module.
- **Example 4:** set  $x = 0$  even in a module where  $x \neq 0$  but  $x^2 = 0$ , so the nilpotent  $x$ -part disappears.

## 8 Final universal picture

The construction

$$(A/I) \otimes_A M$$

means that we are viewing the module  $M$  after passing from the ring  $A$  to the quotient ring  $A/I$ .

Equivalently:

- coefficients are reduced modulo  $I$ ,
- every element of  $I$  acts as zero,
- all elements of  $IM$  vanish,
- the remaining information is exactly the quotient module  $M/IM$ .

So the identity

$$(A/I) \otimes_A M \cong M/IM$$

is the precise algebraic statement that tensoring with  $A/I$  is the same as reducing the module  $M$  modulo the action of the ideal  $I$ .

### Problem 5 — Zero divisors, localization, and associated primes via $D(A)$

**Exact statement.** In a ring  $A$ , let  $D(A)$  denote the set of prime ideals  $\mathfrak{p}$  which satisfy the following condition: there exists an  $a \in A$  such that  $\mathfrak{p}$  is minimal in  $V(\text{Ann}(a))$ .

- (a) Show that  $x$  is a zero divisor in  $A$  if and only if  $x \in \mathfrak{p}$  for some  $\mathfrak{p} \in D(A)$ .
- (b) Let  $S$  be a multiplicatively closed subset of  $A$ . Under the canonical identification of  $\text{Spec } S^{-1}A$  with a subset of  $\text{Spec } A$ , show that

$$D(S^{-1}A) = D(A) \cap \text{Spec } S^{-1}A.$$

- (c) If the 0 ideal has a primary decomposition, show that  $D(A)$  is the set of associated primes of  $A$ .

**Solution.**

- (a) First suppose that  $x$  is a zero divisor. Then there exists  $a \neq 0$  such that

$$xa = 0.$$

Hence  $x \in \text{Ann}(a)$ . Choose a prime ideal  $\mathfrak{p}$  minimal over  $\text{Ann}(a)$ . Then  $\mathfrak{p} \in D(A)$  by definition, and  $\text{Ann}(a) \subseteq \mathfrak{p}$ , so  $x \in \mathfrak{p}$ .

Conversely, suppose  $x \in \mathfrak{p}$  for some  $\mathfrak{p} \in D(A)$ . Then there exists  $a \in A$  such that  $\mathfrak{p}$  is minimal over  $\text{Ann}(a)$ . Let

$$I = \text{Ann}(a), \quad B = A/I.$$

Then  $\mathfrak{q} := \mathfrak{p}/I$  is a minimal prime of  $B$ , and  $\bar{x} \in \mathfrak{q}$ .



Localizing at  $\mathfrak{q}$ , the ring  $B_{\mathfrak{q}}$  has only one prime ideal, namely  $\mathfrak{q}B_{\mathfrak{q}}$ . Hence every element of  $\mathfrak{q}B_{\mathfrak{q}}$  is nilpotent. In particular,  $\bar{x}/1$  is nilpotent in  $B_{\mathfrak{q}}$ . Thus there exist  $s \notin \mathfrak{q}$  and  $n \geq 1$  such that

$$s\bar{x}^n = 0 \quad \text{in } B.$$

Choose  $n$  minimal with this property. Then

$$\bar{b} := s\bar{x}^{n-1} \neq 0, \quad \bar{x}\bar{b} = 0.$$

So  $\bar{x}$  is a zero divisor in  $B$ . Lift  $\bar{b}$  to some  $b \in A$ . Since  $\bar{b} \neq 0$ , we have  $b \notin I = \text{Ann}(a)$ , so  $ba \neq 0$ . Also  $\bar{x}\bar{b} = 0$  means  $xb \in I$ , hence

$$x(ba) = (xb)a = 0.$$

Therefore  $x$  is a zero divisor in  $A$ .

Thus

$$x \text{ is a zero divisor in } A \iff x \in \mathfrak{p} \text{ for some } \mathfrak{p} \in D(A).$$

**(b)** Identify  $\text{Spec } S^{-1}A$  with the set of prime ideals  $\mathfrak{p} \subset A$  such that  $\mathfrak{p} \cap S = \emptyset$ .

First let  $\mathfrak{q} \in D(S^{-1}A)$ . Then there exists  $a/s \in S^{-1}A$  such that  $\mathfrak{q}$  is minimal over

$$\text{Ann}_{S^{-1}A}(a/s) = S^{-1} \text{Ann}_A(a).$$

Write  $\mathfrak{q} = S^{-1}\mathfrak{p}$ , where  $\mathfrak{p} \cap S = \emptyset$ . We claim that  $\mathfrak{p}$  is minimal over  $\text{Ann}_A(a)$ . Indeed, if  $\mathfrak{r} \subsetneq \mathfrak{p}$  is a prime ideal containing  $\text{Ann}_A(a)$ , then  $\mathfrak{r} \cap S = \emptyset$  because  $\mathfrak{r} \subseteq \mathfrak{p}$ . Hence  $S^{-1}\mathfrak{r}$  is a prime of  $S^{-1}A$  properly contained in  $\mathfrak{q}$  and containing  $S^{-1} \text{Ann}_A(a)$ , contradiction. So  $\mathfrak{p} \in D(A)$ , and  $\mathfrak{p} \in \text{Spec } S^{-1}A$ . Therefore

$$D(S^{-1}A) \subseteq D(A) \cap \text{Spec } S^{-1}A.$$

Conversely, let  $\mathfrak{p} \in D(A) \cap \text{Spec } S^{-1}A$ . Then  $\mathfrak{p} \cap S = \emptyset$ , and there exists  $a \in A$  such that  $\mathfrak{p}$  is minimal over  $\text{Ann}_A(a)$ . Since  $\text{Ann}_A(a) \subseteq \mathfrak{p}$  and  $\mathfrak{p} \cap S = \emptyset$ , the element  $a/1$  is nonzero in  $S^{-1}A$ . Also

$$\text{Ann}_{S^{-1}A}(a/1) = S^{-1} \text{Ann}_A(a).$$

If  $S^{-1}\mathfrak{r} \subsetneq S^{-1}\mathfrak{p}$  is a prime containing  $S^{-1} \text{Ann}_A(a)$ , then  $\mathfrak{r} \subsetneq \mathfrak{p}$  is a prime containing  $\text{Ann}_A(a)$ , contradiction. Hence  $S^{-1}\mathfrak{p} \in D(S^{-1}A)$ . Thus

$$D(A) \cap \text{Spec } S^{-1}A \subseteq D(S^{-1}A).$$

Therefore

$$D(S^{-1}A) = D(A) \cap \text{Spec } S^{-1}A.$$

**(c)** Assume that

$$0 = Q_1 \cap \cdots \cap Q_n$$

is an irredundant primary decomposition, where each  $Q_i$  is  $\mathfrak{p}_i$ -primary. Then

$$\text{Ass}_A(A) = \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}.$$

We show that  $D(A)$  is the same set.

First, let  $\mathfrak{p}_i$  be one of these primes. Since the decomposition is irredundant, there exists

$$a_i \in \bigcap_{j \neq i} Q_j \setminus Q_i.$$

Then

$$\text{Ann}(a_i) = \{r \in A : ra_i = 0\}.$$

Because  $a_i \in Q_j$  for all  $j \neq i$ , the condition  $ra_i = 0$  is equivalent to  $ra_i \in Q_i$ . Hence

$$\text{Ann}(a_i) = (Q_i : a_i).$$

Since  $Q_i$  is  $\mathfrak{p}_i$ -primary and  $a_i \notin Q_i$ , the colon ideal  $(Q_i : a_i)$  is  $\mathfrak{p}_i$ -primary. Therefore

$$\sqrt{\text{Ann}(a_i)} = \mathfrak{p}_i,$$

so  $\mathfrak{p}_i$  is the unique minimal prime over  $\text{Ann}(a_i)$ . Hence  $\mathfrak{p}_i \in D(A)$ .

Thus

$$\text{Ass}_A(A) \subseteq D(A).$$

Conversely, let  $\mathfrak{p} \in D(A)$ . Then  $\mathfrak{p}$  is minimal over  $\text{Ann}(a)$  for some  $a \in A$ . Since

$$0 = \bigcap_{i=1}^n Q_i,$$

we obtain

$$\text{Ann}(a) = \bigcap_{i=1}^n (Q_i : a).$$

For each  $i$ , either  $a \in Q_i$ , in which case  $(Q_i : a) = A$ , or  $a \notin Q_i$ , in which case  $(Q_i : a)$  is  $\mathfrak{p}_i$ -primary. Therefore  $\text{Ann}(a)$  has a primary decomposition whose associated primes are among the  $\mathfrak{p}_i$ . Hence every minimal prime over  $\text{Ann}(a)$ , in particular  $\mathfrak{p}$ , must equal one of the  $\mathfrak{p}_i$ . So

$$D(A) \subseteq \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\} = \text{Ass}_A(A).$$

Therefore

$$D(A) = \text{Ass}_A(A).$$

## Problem 6 — The ideals $S_{\mathfrak{p}}(0)$ and minimal primes

**Exact statement.** For  $\mathfrak{p} \subset A$  a prime ideal, denote by  $S_{\mathfrak{p}}(0)$  the kernel of the canonical homomorphism

$$A \rightarrow A_{\mathfrak{p}}.$$

Prove:

- (a)  $S_{\mathfrak{p}}(0) \subseteq \mathfrak{p}$ .
- (b) If  $\mathfrak{p} \supseteq \mathfrak{p}'$ , then  $S_{\mathfrak{p}}(0) \subseteq S_{\mathfrak{p}'}(0)$ .
- (c)  $\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}$  if and only if  $\mathfrak{p}$  is a minimal prime ideal of  $A$ .

- (d)  $\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0$ , where  $D(A)$  is the set defined in the previous exercise.

**Solution.**

Recall that

$$S_{\mathfrak{p}}(0) = \{a \in A : \exists s \notin \mathfrak{p} \text{ such that } sa = 0\}.$$

(a) Let  $a \in S_{\mathfrak{p}}(0)$ . Then  $sa = 0$  for some  $s \notin \mathfrak{p}$ . Since  $0 \in \mathfrak{p}$ , we have  $sa \in \mathfrak{p}$ . Because  $\mathfrak{p}$  is prime and  $s \notin \mathfrak{p}$ , it follows that  $a \in \mathfrak{p}$ . Hence

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{p}.$$

(b) Suppose  $\mathfrak{p} \supseteq \mathfrak{p}'$ , and let  $a \in S_{\mathfrak{p}}(0)$ . Then there exists  $s \notin \mathfrak{p}$  such that  $sa = 0$ . Since  $\mathfrak{p}' \subseteq \mathfrak{p}$ , we also have  $s \notin \mathfrak{p}'$ . Thus  $a \in S_{\mathfrak{p}'}(0)$ . Therefore

$$S_{\mathfrak{p}}(0) \subseteq S_{\mathfrak{p}'}(0).$$

(c) First assume that  $\mathfrak{p}$  is a minimal prime ideal of  $A$ . By part (a),

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{p},$$

so

$$\sqrt{S_{\mathfrak{p}}(0)} \subseteq \mathfrak{p}.$$

Now let  $x \in \mathfrak{p}$ . In the localized ring  $A_{\mathfrak{p}}$ , the prime ideals correspond to prime ideals of  $A$  contained in  $\mathfrak{p}$ . Since  $\mathfrak{p}$  is minimal, the only prime ideal of  $A_{\mathfrak{p}}$  is  $\mathfrak{p}A_{\mathfrak{p}}$ . Therefore the nilradical of  $A_{\mathfrak{p}}$  is  $\mathfrak{p}A_{\mathfrak{p}}$ , so  $x/1$  is nilpotent in  $A_{\mathfrak{p}}$ . Hence for some  $n \geq 1$ ,

$$(x/1)^n = 0,$$

which means  $x^n \in S_{\mathfrak{p}}(0)$ . Thus  $x \in \sqrt{S_{\mathfrak{p}}(0)}$ , and so

$$\mathfrak{p} \subseteq \sqrt{S_{\mathfrak{p}}(0)}.$$

Therefore

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}.$$

Conversely, suppose

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}.$$

If  $\mathfrak{p}$  were not minimal, there would exist a prime ideal  $\mathfrak{q} \subsetneq \mathfrak{p}$ . We claim that

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{q}.$$

Indeed, if  $a \in S_{\mathfrak{p}}(0)$ , then  $sa = 0$  for some  $s \notin \mathfrak{p}$ . Since  $\mathfrak{q} \subseteq \mathfrak{p}$ , also  $s \notin \mathfrak{q}$ . Because  $\mathfrak{q}$  is prime and  $sa = 0 \in \mathfrak{q}$ , it follows that  $a \in \mathfrak{q}$ . Therefore

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{q},$$

hence

$$\sqrt{S_{\mathfrak{p}}(0)} \subseteq \mathfrak{q} \subsetneq \mathfrak{p},$$

contradiction. Thus  $\mathfrak{p}$  must be minimal.

So

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p} \iff \mathfrak{p} \text{ is a minimal prime ideal of } A.$$

(d) We prove that

$$\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0.$$

Let  $x \neq 0$ . Choose a prime ideal  $\mathfrak{p}$  minimal over  $\text{Ann}(x)$ . Then  $\mathfrak{p} \in D(A)$  by definition. We claim that  $x \notin S_{\mathfrak{p}}(0)$ . Otherwise there would exist  $s \notin \mathfrak{p}$  such that

$$sx = 0.$$

Then  $s \in \text{Ann}(x) \subseteq \mathfrak{p}$ , contradiction. Hence  $x \notin S_{\mathfrak{p}}(0)$ .

Therefore no nonzero element belongs to every  $S_{\mathfrak{p}}(0)$ , and we conclude that

$$\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0.$$

## Problem 7 — A local-global criterion for Noetherianity

**Exact statement.** Let  $A$  be a ring such that

- (1) for every maximal ideal  $\mathfrak{m}$  of  $A$ , the local ring  $A_{\mathfrak{m}}$  is Noetherian;
- (2) for each non-zero  $x \in A$ , the set of maximal ideals of  $A$  which contain  $x$  is finite.

Show that  $A$  is Noetherian.

**Solution.**

We show that every ideal of  $A$  is finitely generated.

Let  $I \subseteq A$  be an ideal. If  $I = 0$ , there is nothing to prove. So assume  $I \neq 0$ , and choose  $0 \neq x \in I$ .

By assumption (2), only finitely many maximal ideals contain  $x$ . Denote them by

$$\mathfrak{m}_1, \dots, \mathfrak{m}_r.$$

For each  $i$ , the ideal  $I_{\mathfrak{m}_i}$  of the Noetherian ring  $A_{\mathfrak{m}_i}$  is finitely generated. Hence there exist elements

$$a_{i1}, \dots, a_{in_i} \in I$$

such that

$$I_{\mathfrak{m}_i} = (a_{i1}, \dots, a_{in_i})_{\mathfrak{m}_i}.$$

Now define

$$J = (x, a_{11}, \dots, a_{1n_1}, \dots, a_{r1}, \dots, a_{rn_r}) \subseteq I.$$

This is a finitely generated ideal. We claim that  $I = J$ .

Let  $\mathfrak{m}$  be any maximal ideal of  $A$ .

If  $x \notin \mathfrak{m}$ , then  $x/1$  is a unit in  $A_{\mathfrak{m}}$ . Since  $x \in J \subseteq I$ , both  $J_{\mathfrak{m}}$  and  $I_{\mathfrak{m}}$  contain a unit, hence

$$J_{\mathfrak{m}} = A_{\mathfrak{m}} = I_{\mathfrak{m}}.$$

If  $x \in \mathfrak{m}$ , then  $\mathfrak{m} = \mathfrak{m}_i$  for some  $i$ , and by construction

$$J_{\mathfrak{m}} = I_{\mathfrak{m}}.$$

Thus

$$J_{\mathfrak{m}} = I_{\mathfrak{m}} \quad \text{for every maximal ideal } \mathfrak{m}.$$

Consider the quotient module  $I/J$ . For every maximal ideal  $\mathfrak{m}$ ,

$$(I/J)_{\mathfrak{m}} = 0.$$

A module is zero if and only if all its localizations at maximal ideals are zero. Therefore  $I/J = 0$ , so  $I = J$ .

Hence every ideal of  $A$  is finitely generated. Therefore  $A$  is Noetherian.